

Introduction to Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Deep Learning

Deep Learning uses neural networks with multiple hidden layers to process complex data.

Applications:

- Image Recognition
- Speech Recognition
- Chatbots
- Recommendation Systems

Natural Language Processing (NLP)

NLP allows machines to understand and generate human language.

Examples:

- ChatGPT
- Google Translate
- Voice Assistants

Retrieval-Augmented Generation (RAG)

RAG combines retrieval systems with Large Language Models (LLMs).

RAG Workflow:

1. User asks a query
2. Relevant documents are retrieved from a vector database
3. Retrieved content is passed to the LLM
4. LLM generates a context-aware answer

Components of a RAG System:

- Document Loader
- Text Splitter
- Embedding Model
- Vector Database
- Retriever
- LLM

Popular Vector Databases:

- ChromaDB
- FAISS
- Pinecone
- Weaviate

Popular Embedding Models:

- all-MiniLM-L6-v2
- BAAI/bge-small-en-v1.5

Popular LLM Providers:

- OpenAI
- Mistral AI
- Anthropic
- Google Gemini

Benefits of RAG:

- Reduces hallucinations
- Uses custom datasets
- Provides context-aware answers
- Improves accuracy